NRES 746, Week 11, Lecture 2: Two Cores from One Tree

Name: _____

Instructions: Work alone (about 5 minutes), then compare with a neighbor. Keep laptops closed.

You want the mean annual ring width of the pinyon pines in a stand. Each increment core gives one measurement X with variance $\sigma^2 = 0.04$ mm^2 ($\sigma = 0.2$ mm). Some rules for variances and covariances that may come in handy (a and b are constants):

$$\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X) \qquad \mathrm{Cov}(aX, bY) = ab\,\mathrm{Cov}(X, Y)$$

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y)$$

where $\mathrm{Cov}(X, Y) = \rho\,\sigma_X\sigma_Y$ and ρ is the correlation between X and Y.

You average two cores, $M = \frac{1}{2}(X_1 + X_2)$. Taking the constant out first,

$$\mathrm{Var}(M) = \tfrac{1}{4}\,\mathrm{Var}(X_1 + X_2) = \tfrac{1}{4}\big[\mathrm{Var}(X_1) + \mathrm{Var}(X_2) + 2\,\mathrm{Cov}(X_1, X_2)\big]$$

1a. If the two cores come from different trees ($\rho = 0$), $\mathrm{Var}(M) =$ _____ mm^2.

1b. If both cores come from the same tree ($\rho = 0.8$), $\mathrm{Var}(M) =$ _____ mm^2.